

# Business Case: Target SQL

This business case has data of 100k orders from 2016 to 2018 made at Target, Brazil. It is Americas leading retailer business chain.

Data is available in 8 tables, which gives information about orders from different dimensions like status of order, payment details, location and time of the order , customer who made the purchase, items in the order, product details, seller information of the products, order reviews etc.

## Analysis

### 1. Initial exploration of dataset

1.1 Show all the tables and all the columns present in each table along with its data type.

Query:

```
SELECT
    table_schema,
    table_name,
    column_name,
    data_type
FROM
    `target`.INFORMATION_SCHEMA.COLUMNS;
```

Result:

Query results

 SAVE

JOB INFORMATIONRESULTSJSONEXECUTION DETAILSEXECUTION GRAPHPREVIEW

| Row | table_schema | table_name  | column_name            | data_type |
|-----|--------------|-------------|------------------------|-----------|
| 1   | target       | order_items | order_id               | STRING    |
| 2   | target       | order_items | order_item_id          | INT64     |
| 3   | target       | order_items | product_id             | STRING    |
| 4   | target       | order_items | seller_id              | STRING    |
| 5   | target       | order_items | shipping_limit_date    | TIMESTAMP |
| 6   | target       | order_items | price                  | FLOAT64   |
| 7   | target       | order_items | freight_value          | FLOAT64   |
| 8   | target       | sellers     | seller_id              | STRING    |
| 9   | target       | sellers     | seller_zip_code_prefix | INT64     |
| 10  | target       | sellers     | seller_city            | STRING    |

## 1.2 For which time period the data is given.

Query:

```
SELECT
  MIN(DATE(order_purchase_timestamp)) AS first_date,
  MAX(DATE(order_purchase_timestamp)) AS last_date
FROM
  `target.orders`;
```

Result:

| Query results   |              |             |      |
|-----------------|--------------|-------------|------|
| JOB INFORMATION |              | RESULTS     | JSON |
| Row             | first_date ▼ | last_date ▼ |      |
| 1               | 2016-09-04   | 2018-10-17  |      |

## 1.3 From which Cities and States , orders were placed during the given period.

Query:

```
SELECT
  DISTINCT c.customer_state , c.customer_city
FROM
  `target.customers` c
RIGHT JOIN
  `target.orders` o
USING
  (customer_id)
ORDER BY
  c.customer_state , c.customer_city;
```

Result:

| Row | customer_state ▼ | customer_city ▼  |
|-----|------------------|------------------|
| 1   | AC               | brasileia        |
| 2   | AC               | cruzeiro do sul  |
| 3   | AC               | epitaciolandia   |
| 4   | AC               | manoel urbano    |
| 5   | AC               | porto acre       |
| 6   | AC               | rio branco       |
| 7   | AC               | senador guiomard |
| 8   | AC               | xapuri           |
| 9   | AL               | agua branca      |
| 10  | AL               | anadia           |

#### 1.4 What is distribution of total orders as per their status?

Query:

```
SELECT
    order_status,
    COUNT(*) AS orders_count
FROM
    `target.orders`
GROUP BY
    order_status
ORDER BY
    orders_count DESC;
```

Result:

##### Query results

| JOB INFORMATION |                | RESULTS        | JSON | EXT |
|-----------------|----------------|----------------|------|-----|
| Row             | order_status ▼ | orders_count ▼ |      |     |
| 1               | delivered      | 96478          |      |     |
| 2               | shipped        | 1107           |      |     |
| 3               | canceled       | 625            |      |     |
| 4               | unavailable    | 609            |      |     |
| 5               | invoiced       | 314            |      |     |
| 6               | processing     | 301            |      |     |
| 7               | created        | 5              |      |     |
| 8               | approved       | 2              |      |     |

#### 1.5 How is the order value of each order is calculated in payment table, Is it addition of price and freight for each item in the order.

*(Used order items and payment table)*

```
SELECT
    order_id, total_price, total_freight,
    total_price + total_freight AS total_order_value
FROM ( SELECT
    order_id,
    SUM(price) AS total_price,
    SUM(freight_value) AS total_freight,
    COUNT(*) AS total_items
FROM
    `target.order_items`
```

```

WHERE
  order_id = "95d6357ffe41aa6d2998852a710c70a0"
GROUP BY
  order_id) t1;

```

Result:

| Query results   |                                  |             |               |                   |                                      |
|-----------------|----------------------------------|-------------|---------------|-------------------|--------------------------------------|
| JOB INFORMATION |                                  | RESULTS     | JSON          | EXECUTION DETAILS | EXECUTION GRAPH <span>PREVIEW</span> |
| Row             | order_id                         | total_price | total_freight | total_order_value |                                      |
| 1               | 95d6357ffe41aa6d2998852a710c70a0 | 17.5        | 91.15         | 108.65            |                                      |

```

SELECT
  order_id, payment_value
FROM
  `target.payments`
WHERE
  order_id = "95d6357ffe41aa6d2998852a710c70a0";

```

| Query results   |                                  |               |               |
|-----------------|----------------------------------|---------------|---------------|
| JOB INFORMATION |                                  | RESULTS       | EXECUTION DET |
| Row             | order_id                         | payment_value |               |
| 1               | 95d6357ffe41aa6d2998852a710c70a0 | 108.65        |               |

## 2. In-depth exploration of dataset

2.1 Is there a growing trend on e-commerce in Brazil? How can we describe a complete scenario?

Query:

```

SELECT
  time_period,
  order_count,

```

```

ROUND((((order_count - LAG(order_count) OVER(ORDER BY t1.YEAR, t1.month)) / LAG(order_count)
OVER(ORDER BY t1.YEAR, t1.month))* 100), 2) AS growth_percent
FROM (
SELECT
  EXTRACT(YEAR
FROM
  order_purchase_timestamp) AS year,
  EXTRACT(MONTH
FROM
  order_purchase_timestamp) AS month,
  FORMAT_DATE('%b %Y', DATE(ORDER_PURCHASE_TIMESTAMP)) AS time_period,
  COUNT(order_id) AS order_count
FROM
  `target.orders`
WHERE
  order_status = "delivered"
GROUP BY
  month, year, time_period) t1
ORDER BY
  year, month;

```

Result:

| Query results   |             |             |                |                   |
|-----------------|-------------|-------------|----------------|-------------------|
| JOB INFORMATION |             | RESULTS     | JSON           | EXECUTION DETAILS |
| Row             | time_period | order_count | growth_percent |                   |
| 1               | Sep 2016    | 1           | null           |                   |
| 2               | Oct 2016    | 265         | 26400.0        |                   |
| 3               | Dec 2016    | 1           | -99.62         |                   |
| 4               | Jan 2017    | 750         | 74900.0        |                   |
| 5               | Feb 2017    | 1653        | 120.4          |                   |
| 6               | Mar 2017    | 2546        | 54.02          |                   |
| 7               | Apr 2017    | 2303        | -9.54          |                   |
| 8               | May 2017    | 3546        | 53.97          |                   |
| 9               | Jun 2017    | 3135        | -11.59         |                   |
| 10              | Jul 2017    | 3872        | 23.51          |                   |

Can we see some seasonality with peaks at specific months?

No, as for some months data is showing orders of only 1 or 2 records and also there is huge spike in orders is seen in different months so we can't comment on seasonality.  
Overall business is in uptrend and sharp spike in orders is seen MoM basis.

## 2.2 What time do Brazilian customers tend to buy (Dawn, Morning, Afternoon or Night)?

Query:

```
SELECT
  temp.purchase_time,
  COUNT(*) AS total_orders
FROM (
  SELECT
    order_id,
    CASE
      WHEN TIME(order_purchase_timestamp) BETWEEN "00:00:00" AND "07:00:00" THEN "Dawn"
      WHEN TIME(order_purchase_timestamp) BETWEEN "07:00:01" AND "12:00:00" THEN "Morning"
      WHEN TIME(order_purchase_timestamp) BETWEEN "12:00:01" AND "18:00:00" THEN "Afternoon"
      WHEN TIME(order_purchase_timestamp) BETWEEN "18:00:01" AND "23:59:59" THEN "Night"
    END
    AS purchase_time
  FROM
    `target.orders`) TEMP
GROUP BY
  temp.purchase_time
ORDER BY
  total_orders DESC;
```

Result:

| Query results   |               |              |                |
|-----------------|---------------|--------------|----------------|
| JOB INFORMATION |               | RESULTS      | JSON           |
| Row             | purchase_time | total_orders | EXECUTION TIME |
| 1               | Afternoon     | 38365        |                |
| 2               | Night         | 34096        |                |
| 3               | Morning       | 21738        |                |
| 4               | Dawn          | 5242         |                |

**Brazilian customers usually tend to buy in afternoon and night.**

### 3. Evolution of E-commerce orders in the Brazil region:

#### 3.1 Get month on month orders by states

Query:

```
SELECT
  state , time_period , total_orders,
  LAG(total_orders) OVER(PARTITION BY state ORDER BY year, month ) AS prev_month_orders_count,
  ROUND(((total_orders - LAG(total_orders) OVER(PARTITION BY state ORDER BY year, month )) /
LAG(total_orders) OVER(PARTITION BY state ORDER BY year, month))* 100,2) AS MoM_percent_growth
FROM (
  SELECT
    state,
    time_period,
    year,
    month,
    COUNT(*) AS total_orders
  FROM (
    SELECT
      o.order_id, o.order_purchase_timestamp,
      EXTRACT(YEAR FROM order_purchase_timestamp) AS year,
      EXTRACT(Month FROM order_purchase_timestamp) AS month,
      FORMAT_DATE('%b %Y', DATE(ORDER_PURCHASE_TIMESTAMP)) AS time_period,
      c.customer_state AS state
    FROM
      `target.orders` o
    JOIN
      `target.customers` c
    USING
      (customer_id)
    ORDER BY
      year, month) t1
  GROUP BY
    state, time_period,year, month) t2;
```

Query results



| JOB INFORMATION |       | RESULTS     | JSON         | EXECUTION DETAILS | EXECUTION GRAPH    | PREVIEW |
|-----------------|-------|-------------|--------------|-------------------|--------------------|---------|
| Row             | state | time_period | total_orders | prev_month_orders | MoM_percent_growth |         |
| 1               | AL    | Oct 2016    | 2            | null              | null               |         |
| 2               | AL    | Jan 2017    | 2            | 2                 | 0.0                |         |
| 3               | AL    | Feb 2017    | 12           | 2                 | 500.0              |         |
| 4               | AL    | Mar 2017    | 10           | 12                | -16.67             |         |
| 5               | AL    | Apr 2017    | 23           | 10                | 130.0              |         |
| 6               | AL    | May 2017    | 27           | 23                | 17.39              |         |
| 7               | AL    | Jun 2017    | 10           | 27                | -62.96             |         |
| 8               | AL    | Jul 2017    | 17           | 10                | 70.0               |         |
| 9               | AL    | Aug 2017    | 18           | 17                | 5.88               |         |
| 10              | AL    | Sep 2017    | 20           | 18                | 11.11              |         |

### 3.2 Distribution of customers across the states in Brazil

Query:

```
SELECT
  customer_state AS state,
  COUNT(*) AS total_customers
FROM
  `target.customers`
GROUP BY
  customer_state
ORDER BY
  total_customers DESC;
```

Result:

#### Query results

| JOB INFORMATION |       | RESULTS         | JSON | EXE |
|-----------------|-------|-----------------|------|-----|
| Row             | state | total_customers |      |     |
| 1               | SP    | 41746           |      |     |
| 2               | RJ    | 12852           |      |     |
| 3               | MG    | 11635           |      |     |
| 4               | RS    | 5466            |      |     |
| 5               | PR    | 5045            |      |     |
| 6               | SC    | 3637            |      |     |
| 7               | BA    | 3380            |      |     |
| 8               | DF    | 2140            |      |     |
| 9               | ES    | 2033            |      |     |
| 10              | GO    | 2020            |      |     |

## 4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

4.1 Get % increase in cost of orders from 2017 to 2018 (include months between Jan to Aug only) - You can use “payment\_value” column in payments table

Query:

```
SELECT
  *,
  COALESCE((ROUND(((total_orders_value - LAG(total_orders_value) OVER(ORDER BY
year)))/LAG(total_orders_value) OVER(ORDER BY year))* 100, 2)), 0) AS percent_increase_YOY
FROM (
```



```

SELECT
  year,
  ROUND(SUM(payment_value), 2) AS total_orders_value
FROM (
  SELECT
    o.order_id,
    o.order_purchase_timestamp,
    EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month,
    EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year,
    p.payment_value
  FROM
    `target.orders` o
  JOIN
    `target.payments` p
  USING
    (order_id)
  WHERE
    o.order_status = "delivered")t1
WHERE
  month BETWEEN 1
  AND 8
GROUP BY
  year) t1
ORDER BY
  Year;

```

## Query results

| JOB INFORMATION |        | RESULTS              | JSON | EXECUTION DETAILS      |  |
|-----------------|--------|----------------------|------|------------------------|--|
| Row             | year ▼ | total_orders_value ▼ |      | percent_increase_YOY ▼ |  |
| 1               | 2017   | 3473862.76           |      | 0.0                    |  |
| 2               | 2018   | 8452975.2            |      | 143.33                 |  |

Comparison on Monthly total order value:

Query:

```

SELECT
  t2.Month_n_Year,
  t2.total_orders_value,
  ROUND(((t2.total_orders_value - LAG(total_orders_value) OVER(PARTITION BY month ORDER BY
year))/LAG(total_orders_value) OVER(PARTITION BY month ORDER BY year))*100, 2) AS
percent_increase
FROM (
  SELECT
    month,
    year,

```

```

Month_n_Year,
ROUND(SUM(payment_value)) AS total_orders_value
FROM (
SELECT
o.order_id,
o.order_purchase_timestamp,
EXTRACT(month FROM o.order_purchase_timestamp ) AS month,
EXTRACT(year FROM o.order_purchase_timestamp ) AS year,
FORMAT_DATE('%b %Y', DATE(ORDER_PURCHASE_TIMESTAMP)) AS Month_n_Year,
p.payment_value
FROM
`target.orders` o
JOIN
`target.payments` p
USING
(order_id)) t1
WHERE
month BETWEEN 1
AND 8
GROUP BY
month, year, Month_n_Year
ORDER BY
month, year) t2
ORDER BY
month;

```

## Query results

| JOB INFORMATION |              | RESULTS            | JSON             | EXECUTION DETAILS |
|-----------------|--------------|--------------------|------------------|-------------------|
| Row             | Month_n_Year | total_orders_value | percent_increase |                   |
| 1               | Jan 2017     | 138488.0           | null             |                   |
| 2               | Jan 2018     | 1115004.0          | 705.13           |                   |
| 3               | Feb 2017     | 291908.0           | null             |                   |
| 4               | Feb 2018     | 992463.0           | 239.99           |                   |
| 5               | Mar 2017     | 449864.0           | null             |                   |
| 6               | Mar 2018     | 1159652.0          | 157.78           |                   |
| 7               | Apr 2017     | 417788.0           | null             |                   |
| 8               | Apr 2018     | 1160785.0          | 177.84           |                   |
| 9               | May 2017     | 592919.0           | null             |                   |
| 10              | May 2018     | 1153982.0          | 94.63            |                   |

## 4.2 Mean & Sum of price and freight value by customer state

Query:

```
SELECT
  c.customer_state,
  ROUND(SUM(oi.price)) total_price,
  ROUND(AVG(oi.price)) avg_price,
  ROUND(SUM(oi.freight_value)) total_freight,
  ROUND(AVG(oi.freight_value)) avg_freight
FROM
  `target.order_items` oi
JOIN
  `target.orders` o
USING
  (order_id)
JOIN
  `target.customers` c
ON
  c.customer_id = o.customer_id
GROUP BY
  c.customer_state;
```

### Query results

| JOB INFORMATION |                | RESULTS     | JSON      | EXECUTION DETAILS |             | EXECUTION GRAPH | PREVIEW |
|-----------------|----------------|-------------|-----------|-------------------|-------------|-----------------|---------|
| Row             | customer_state | total_price | avg_price | total_freight     | avg_freight |                 |         |
| 1               | MT             | 156454.0    | 148.0     | 29715.0           | 28.0        |                 |         |
| 2               | MA             | 119648.0    | 145.0     | 31524.0           | 38.0        |                 |         |
| 3               | AL             | 80315.0     | 181.0     | 15915.0           | 36.0        |                 |         |
| 4               | SP             | 5202955.0   | 110.0     | 718723.0          | 15.0        |                 |         |
| 5               | MG             | 1585308.0   | 121.0     | 270853.0          | 21.0        |                 |         |
| 6               | PE             | 262788.0    | 146.0     | 59450.0           | 33.0        |                 |         |
| 7               | RJ             | 1824093.0   | 125.0     | 305589.0          | 21.0        |                 |         |
| 8               | DF             | 302604.0    | 126.0     | 50625.0           | 21.0        |                 |         |
| 9               | RS             | 750304.0    | 120.0     | 135523.0          | 22.0        |                 |         |
| 10              | SE             | 58921.0     | 153.0     | 14111.0           | 37.0        |                 |         |

## 5. Analysis on sales, freight and delivery time

### 5.1 Calculate days between purchasing, delivering and estimated delivery

Query:

```
SELECT
  order_id,
  TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp, day) AS
  actual_delivery_time_in_days ,
  TIMESTAMP_DIFF(order_estimated_delivery_date, order_purchase_timestamp, day) AS
  estimated_delivery_time_in_days
FROM
  `target.orders`
WHERE
  order_status = "delivered";
```

#### Query results

| JOB INFORMATION |                               | RESULTS                      | JSON                            | EXECUTION DETAILS | EXECUTION GRAPH | PREVIEW |
|-----------------|-------------------------------|------------------------------|---------------------------------|-------------------|-----------------|---------|
| Row             | order_id                      | actual_delivery_time_in_days | estimated_delivery_time_in_days |                   |                 |         |
| 1               | 635c894d068ac37e6e03dc54e...  | 30                           | 32                              |                   |                 |         |
| 2               | 3b97562c3aee8bdedcb5c2e45...  | 32                           | 33                              |                   |                 |         |
| 3               | 68f47f50f04c4cb6774570cfde... | 29                           | 31                              |                   |                 |         |
| 4               | 276e9ec344d3bf029ff83a161c... | 43                           | 39                              |                   |                 |         |
| 5               | 54e1a3c2b97fb0809da548a59...  | 40                           | 36                              |                   |                 |         |
| 6               | fd04fa4105ee8045f6a0139ca5... | 37                           | 35                              |                   |                 |         |
| 7               | 302bb8109d097a9fc6e9cefc5...  | 33                           | 28                              |                   |                 |         |
| 8               | 66057d37308e787052a32828...   | 38                           | 32                              |                   |                 |         |
| 9               | 19135c945c554eebfd7576c73...  | 36                           | 33                              |                   |                 |         |
| 10              | 4493e45e7ca1084efcd38ddeb...  | 34                           | 33                              |                   |                 |         |

### 5.2 Find time\_to\_delivery & diff\_estimated\_delivery. Formula for the same given below:

time\_to\_delivery = order\_delivered\_customer\_date-order\_purchase\_timestamp

diff\_estimated\_delivery = order\_estimated\_delivery\_date-order\_delivered\_customer\_date

```
SELECT
```

```

    TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp, day) AS
time_to_delivery,
    TIMESTAMP_DIFF(order_estimated_delivery_date, order_delivered_customer_date, day) AS
diff_estimated_delivery
FROM
    `target.orders`
WHERE
    order_status = "delivered";

```

## Query results

| JOB INFORMATION |                  | RESULTS                 | JSON | EXECUT |
|-----------------|------------------|-------------------------|------|--------|
| Row             | time_to_delivery | diff_estimated_delivery |      |        |
| 1               | 30               | 1                       |      |        |
| 2               | 32               | 0                       |      |        |
| 3               | 29               | 1                       |      |        |
| 4               | 43               | -4                      |      |        |
| 5               | 40               | -4                      |      |        |
| 6               | 37               | -1                      |      |        |
| 7               | 33               | -5                      |      |        |
| 8               | 38               | -6                      |      |        |
| 9               | 36               | -2                      |      |        |
| 10              | 34               | 0                       |      |        |

## 5.3 Group data by state, take mean of freight\_value, time\_to\_delivery, diff\_estimated\_delivery

```

SELECT
    c.customer_state,
    ROUND(AVG(oi.freight_value),2) AS avg_freight_value,
    ROUND(AVG(o.time_to_delivery),2) AS avg_time_to_delivery,
    ROUND(AVG(o.diff_estimated_delivery),2) AS avg_diff_estimated_delivery
FROM
    `target.order_items` oi JOIN
    ( SELECT
        order_id, customer_id,
        TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp, day) AS
time_to_delivery,
        TIMESTAMP_DIFF(order_estimated_delivery_date, order_delivered_customer_date, day) AS
diff_estimated_delivery
    FROM
        `target.orders`

```

```

WHERE
    order_status = "delivered") o
USING
    (order_id)
JOIN
    `target.customers` c
USING
    (customer_id)
GROUP BY
    customer_state;
Query results

```

| JOB INFORMATION |                | RESULTS           | JSON                 | EXECUTION DETAILS           | EXECUTION GRAPH | PREVIEW |
|-----------------|----------------|-------------------|----------------------|-----------------------------|-----------------|---------|
| Row             | customer_state | avg_freight_value | avg_time_to_delivery | avg_diff_estimated_delivery |                 |         |
| 1               | SP             | 15.12             | 8.26                 | 10.26                       |                 |         |
| 2               | RJ             | 20.91             | 14.69                | 11.14                       |                 |         |
| 3               | PR             | 20.47             | 11.48                | 12.53                       |                 |         |
| 4               | SC             | 21.51             | 14.52                | 10.66                       |                 |         |
| 5               | DF             | 21.07             | 12.5                 | 11.27                       |                 |         |
| 6               | MG             | 20.63             | 11.51                | 12.4                        |                 |         |
| 7               | PA             | 35.63             | 23.3                 | 13.37                       |                 |         |
| 8               | BA             | 26.49             | 18.77                | 10.12                       |                 |         |
| 9               | GO             | 22.56             | 14.95                | 11.37                       |                 |         |
| 10              | RS             | 21.61             | 14.71                | 13.2                        |                 |         |

5.4 Sort the data to get the following: Top 5 states with highest/lowest average freight value - sort in desc/asc limit 5

States with Highest Avg freight value

```

SELECT
    customer_state
FROM (
    SELECT
        customer_state,
        ROUND(AVG(freight_value),2) AS avg_freight
    FROM
        `target.order_items` oi
    JOIN
        `target.orders` o
    USING
        (order_id)
    JOIN
        `target.customers` c
    USING
        (customer_id)
    GROUP BY

```

```

        customer_state
ORDER BY
        avg_freight DESC) t1
LIMIT 5;

```

| JOB INFORMATION |                | RESULTS |
|-----------------|----------------|---------|
| Row             | customer_state |         |
| 1               | RR             |         |
| 2               | PB             |         |
| 3               | RO             |         |
| 4               | AC             |         |
| 5               | PI             |         |

States with lowest Avg freight value

```

SELECT
    customer_state
FROM (
    SELECT
        customer_state,
        ROUND(AVG(freight_value),2) AS avg_freight
    FROM
        `target.order_items` oi
    JOIN
        `target.orders` o
    USING
        (order_id)
    JOIN
        `target.customers` c
    USING
        (customer_id)
    GROUP BY
        customer_state
    ORDER BY
        avg_freight ) t1
LIMIT 5;

```

## Query results

| JOB INFORMATION |                | RESULTS |
|-----------------|----------------|---------|
| Row             | customer_state |         |
| 1               | SP             |         |
| 2               | PR             |         |
| 3               | MG             |         |
| 4               | RJ             |         |
| 5               | DF             |         |

### 5.5 Top 5 states with highest/lowest average time to delivery

States with highest average time to delivery

```
SELECT
  customer_state
FROM (
  SELECT
    customer_state,
    ROUND(AVG(time_to_delivery),2) AS avg_delivery_time
  FROM (
    SELECT
      customer_state,
      TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp, day) AS
time_to_delivery,
    FROM
      `target.orders` o
    JOIN
      `target.customers` c
    USING
      (customer_id)
    WHERE
      order_status = "delivered") t1
  GROUP BY
    customer_state
  ORDER BY
    avg_delivery_time DESC) t2
LIMIT 5;
```



## Query results

| JOB INFORMATION |                | RESULTS |
|-----------------|----------------|---------|
| Row             | customer_state |         |
| 1               | RR             |         |
| 2               | AP             |         |
| 3               | AM             |         |
| 4               | AL             |         |
| 5               | PA             |         |

States with lowest average time to delivery

```
SELECT
  customer_state
FROM (
  SELECT
    customer_state,
    ROUND(AVG(time_to_delivery),2) AS avg_delivery_time
  FROM (
    SELECT
      customer_state,
      TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp, day) AS
time_to_delivery,
    FROM
      `target.orders` o
    JOIN
      `target.customers` c
    USING
      (customer_id)
    WHERE
      order_status = "delivered") t1
  GROUP BY
    customer_state
  ORDER BY
    avg_delivery_time ) t2
LIMIT 5;
```

## Query results

| JOB INFORMATION |                | RESULTS |
|-----------------|----------------|---------|
| Row             | customer_state |         |
| 1               | SP             |         |
| 2               | PR             |         |
| 3               | MG             |         |
| 4               | DF             |         |
| 5               | SC             |         |

### 5.6 Top 5 states with really fast delivery compared to estimated date

```
SELECT
  customer_state,
  ROUND(AVG(estimated_delivery_time - actual_delivery_time), 2) AS delivery_time_difference
FROM (
  SELECT
    o.order_id,
    c.customer_state,
    TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp, day) AS
actual_delivery_time,
    TIMESTAMP_DIFF(order_estimated_delivery_date, order_purchase_timestamp, day) AS
estimated_delivery_time
  FROM
    `target.orders` o
  JOIN
    `target.customers` c
  USING
    (customer_id)
  WHERE
    order_status = "delivered") t1
GROUP BY
  customer_state
ORDER BY
  delivery_time_difference DESC
LIMIT 5;
```

## Query results

| JOB INFORMATION |                | RESULTS                  | JSON | EXECUTION I |
|-----------------|----------------|--------------------------|------|-------------|
| Row             | customer_state | delivery_time_difference |      |             |
| 1               | AC             | 20.09                    |      |             |
| 2               | RO             | 19.47                    |      |             |
| 3               | AP             | 19.13                    |      |             |
| 4               | AM             | 18.94                    |      |             |
| 5               | RR             | 16.66                    |      |             |

states with not so fast delivery compared to estimated date

```
SELECT
  customer_state,
  ROUND(AVG(estimated_delivery_time - actual_delivery_time), 2) AS delivery_time_difference
FROM (
  SELECT
    o.order_id,
    c.customer_state,
    TIMESTAMP_DIFF(order_delivered_customer_date, order_purchase_timestamp, day) AS
actual_delivery_time,
    TIMESTAMP_DIFF(order_estimated_delivery_date, order_purchase_timestamp, day) AS
estimated_delivery_time
  FROM
    `target.orders` o
  JOIN
    `target.customers` c
  USING
    (customer_id)
  WHERE
    order_status = "delivered") t1
GROUP BY
  customer_state
ORDER BY
  delivery_time_difference
LIMIT 5;
```

Query results

| JOB INFORMATION |                  | RESULTS                    | JSON | EXECUTION D |
|-----------------|------------------|----------------------------|------|-------------|
| Row             | customer_state ▼ | delivery_time_difference ▼ |      |             |
| 1               | AL               | 8.17                       |      |             |
| 2               | MA               | 8.97                       |      |             |
| 3               | SE               | 9.45                       |      |             |
| 4               | ES               | 9.89                       |      |             |
| 5               | CE               | 10.19                      |      |             |

## 6. Payment type analysis:

### 6.1 Month over Month count of orders for different payment types

```
SELECT
  time_period,
  payment_type,
  COUNT(*) AS total_orders
FROM (
  SELECT
    p.order_id,
    p.payment_type,
    EXTRACT(YEAR FROM order_purchase_timestamp) AS year,
    EXTRACT(Month FROM order_purchase_timestamp) AS month,
    FORMAT_DATE('%b %Y', DATE(ORDER_PURCHASE_TIMESTAMP)) AS time_period
  FROM
    `target.payments` p
  JOIN
    `target.orders` o
  USING
    (order_id)) t1
GROUP BY
  time_period, payment_type, t1.YEAR, t1.month
ORDER BY
  t1.YEAR, t1.month ;
```

### Query results

| JOB INFORMATION |             | RESULTS      | JSON         | EXECUTION DETAILS |
|-----------------|-------------|--------------|--------------|-------------------|
| Row             | time_period | payment_type | total_orders |                   |
| 1               | Sep 2016    | credit_card  | 3            |                   |
| 2               | Oct 2016    | credit_card  | 254          |                   |
| 3               | Oct 2016    | voucher      | 23           |                   |
| 4               | Oct 2016    | debit_card   | 2            |                   |
| 5               | Oct 2016    | UPI          | 63           |                   |
| 6               | Dec 2016    | credit_card  | 1            |                   |
| 7               | Jan 2017    | voucher      | 61           |                   |
| 8               | Jan 2017    | UPI          | 197          |                   |
| 9               | Jan 2017    | credit_card  | 583          |                   |
| 10              | Jan 2017    | debit_card   | 9            |                   |

## 6.2 Count of orders based on the no. of payment installments

```
SELECT
  payment_installments,
  COUNT(*) AS total_orders
FROM
  `target.payments`
GROUP BY
  payment_installments;
```

### Query results

| JOB INFORMATION |                      | RESULTS      | JSON |
|-----------------|----------------------|--------------|------|
| Row             | payment_installments | total_orders |      |
| 1               | 0                    | 2            |      |
| 2               | 1                    | 52546        |      |
| 3               | 2                    | 12413        |      |
| 4               | 3                    | 10461        |      |
| 5               | 4                    | 7098         |      |
| 6               | 5                    | 5239         |      |
| 7               | 6                    | 3920         |      |
| 8               | 7                    | 1626         |      |
| 9               | 8                    | 4268         |      |
| 10              | 9                    | 644          |      |

## 7. Actionable Insights

- Total 609 orders were unavailable and 625 orders were cancelled during the given time period, which makes it to be around 1.2 % of total orders.  
We can reduce this number by studying the reasons behind order cancellation and items unavailability.

### Query results

| JOB INFORMATION |              | RESULTS      | JSON                    | EXECUTION DETAILS | EXECU |
|-----------------|--------------|--------------|-------------------------|-------------------|-------|
| Row             | order_status | orders_count | percent_of_total_orders |                   |       |
| 1               | delivered    | 96478        | 97.02                   |                   |       |
| 2               | shipped      | 1107         | 1.11                    |                   |       |
| 3               | canceled     | 625          | 0.63                    |                   |       |
| 4               | unavailable  | 609          | 0.61                    |                   |       |
| 5               | invoiced     | 314          | 0.32                    |                   |       |
| 6               | processing   | 301          | 0.3                     |                   |       |
| 7               | created      | 5            | 0.01                    |                   |       |
| 8               | approved     | 2            | 0.0                     |                   |       |

- We can see how the orders trajectory is showing very abrupt increase in orders volume with in very short time. Looking at overall trend, it is seen that business is picking up very fast in brazil so company has to be ready with extra workforce. To avoid high risk, it can consider hiring contractual employees.

| Query results   |             |             |                |
|-----------------|-------------|-------------|----------------|
| JOB INFORMATION |             | RESULTS     | JSON           |
| Row             | time_period | order_count | growth_percent |
| 1               | Sep 2016    | 1           | null           |
| 2               | Oct 2016    | 265         | 26400.0        |
| 3               | Dec 2016    | 1           | -99.62         |
| 4               | Jan 2017    | 750         | 74900.0        |
| 5               | Feb 2017    | 1653        | 120.4          |
| 6               | Mar 2017    | 2546        | 54.02          |
| 7               | Apr 2017    | 2303        | -9.54          |
| 8               | May 2017    | 3546        | 53.97          |
| 9               | Jun 2017    | 3135        | -11.59         |
| 10              | Jul 2017    | 3872        | 23.51          |

- Company received low rating for maximum orders in highlighted states; need to study further about the reasons for customer dissatisfaction to such great extent in these states.

This is the query for counting the number of rating in each state.

```

SELECT
*
FROM (
  SELECT
    c.customer_state,
    orv.review_score
  FROM
    `target.order_reviews` orv
  JOIN
    `target.orders` o
  USING
    (order_id)
  JOIN
    `target.customers` c
  USING
    (customer_id)) PIVOT(COUNT(*) FOR review_score IN (1, 2, 3, 4, 5));

```



## Query results

| JOB INFORMATION |                | RESULTS | JSON | EXECUTION DETAILS |      | EXECUTION GRAPH | PREVIEW |
|-----------------|----------------|---------|------|-------------------|------|-----------------|---------|
| Row             | customer_state | _1      | _2   | _3                | _4   | _5              |         |
| 1               | RS             | 560     | 172  | 449               | 1098 | 3204            |         |
| 2               | RJ             | 2183    | 464  | 1050              | 2137 | 6931            |         |
| 3               | PR             | 473     | 156  | 381               | 1009 | 3019            |         |
| 4               | SC             | 413     | 119  | 321               | 712  | 2058            |         |
| 5               | SP             | 4054    | 1211 | 3299              | 7991 | 25135           |         |
| 6               | BA             | 504     | 130  | 337               | 744  | 1642            |         |
| 7               | GO             | 233     | 67   | 191               | 423  | 1110            |         |
| 8               | MG             | 1207    | 339  | 969               | 2259 | 6851            |         |
| 9               | PE             | 221     | 53   | 131               | 322  | 919             |         |
| 10              | RO             | 24      | 15   | 27                | 44   | 142             |         |
| 11              | RN             | 54      | 14   | 39                | 95   | 280             |         |
| 12              | SE             | 63      | 13   | 29                | 67   | 177             |         |
| 13              | MA             | 131     | 31   | 74                | 157  | 353             |         |
| 14              | PA             | 155     | 35   | 96                | 197  | 485             |         |
| 15              | CE             | 210     | 54   | 131               | 263  | 671             |         |
| 16              | ES             | 243     | 57   | 182               | 425  | 1109            |         |

## 8. Recommendations

- As Brazilian customers usually tend to buy in afternoon and night, we can increase staff in during this time frame in order to manage the customers' requests, and services better during this time by reducing workforce of morning and dawn.
- We can see, only 3 state contribute for maximum volume, and rest of the state need to be focused for improving the business.

### Query results

| JOB INFORMATION |                | RESULTS      | JSON | EXECU |
|-----------------|----------------|--------------|------|-------|
| Row             | customer_state | orders_count |      |       |
| 1               | SP             | 41746        |      |       |
| 2               | RJ             | 12852        |      |       |
| 3               | MG             | 11635        |      |       |
| 4               | RS             | 5466         |      |       |
| 5               | PR             | 5045         |      |       |
| 6               | SC             | 3637         |      |       |
| 7               | BA             | 3380         |      |       |
| 8               | DF             | 2140         |      |       |
| 9               | ES             | 2033         |      |       |
| 10              | GO             | 2020         |      |       |
| 11              | PE             | 1652         |      |       |
| 12              | CE             | 1336         |      |       |
| 13              | PA             | 975          |      |       |
| 14              | MT             | 807          |      |       |

- Avg delivery time is quite high for most of those states from where company is receiving quite less volume of orders, detailed study is needed further for checking the other reasons behind such low volume of orders from majority of states. Huge delivery time can be the one of the reason and need to work on it.  
States with highest average delivery time -

## Query results

| JOB INFORMATION |                | RESULTS           | JSON | EXE |
|-----------------|----------------|-------------------|------|-----|
| Row             | customer_state | avg_delivery_time |      |     |
| 1               | RR             | 28.98             |      |     |
| 2               | AP             | 26.73             |      |     |
| 3               | AM             | 25.99             |      |     |
| 4               | AL             | 24.04             |      |     |
| 5               | PA             | 23.32             |      |     |
| 6               | MA             | 21.12             |      |     |
| 7               | SE             | 21.03             |      |     |
| 8               | CE             | 20.82             |      |     |
| 9               | AC             | 20.64             |      |     |
| 10              | PB             | 19.95             |      |     |
| 11              | PI             | 18.99             |      |     |
| 12              | RO             | 18.91             |      |     |
| 13              | BA             | 18.87             |      |     |
| 14              | RN             | 18.82             |      |     |
| 15              | PE             | 17.97             |      |     |
| 16              | MT             | 17.59             |      |     |
| 17              | TO             | 17.23             |      |     |